

Sugandima Weragoda

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OBJECTIVE

I am a highly skilled and innovative researcher with 5+ years of experience in computational physics. I excel in communication and collaboration within team environments and am seeking a meaningful role in which I can explore my passion for science communication and education.

EDUCATION

Ph.D. in Physics, Case Western Reserve University (CWRU), Cleveland, OH	2018-2023 (Expected)
B.Sc. (Hons) in Computational Physics, University of Colombo (UOC), Sri Lanka	2013-2017

RESEARCH EXPERIENCE

Taussig Cancer Center, Cleveland Clinic Foundation (CCF) Cleveland, OH

Visiting Researcher, 2022-Present

- Designed and developed proof of principle prototypes of clinical tools for medical image processing.
- Presented research at AAPM (500+ attendees). (Received travel award)

Project: Deep Learning Based liver Dome Delineation on KV Planar Images for Automatic Liver Dome Gating.

- Developed a proof of principle pipeline for a clinically feasible tool for liver dome delineation on x-ray images.
- The tool is fully automatic and achieves delineations comparable to manual delineation in under one second per image.

Project: Deep Learning Based Automatic Deformable Image Registration for Head and Neck CT.

- Developed a proof of principle pipeline for a clinically feasible tool for Head and Neck CT DIR.
- Developed a deep-learning based model for automatic 3D segmentation and registration of bones in CT which achieves accuracy comparable to the benchmark manual technique.
- Develop a FEA based bio-mechanical model for fast and accurate soft tissue deformation.

MRI Imaging Physics Research Group (CWRU) Cleveland, OH

Doctoral Candidate, 2018-Present

- Mentored undergraduate researchers to complete senior projects in medical image processing.
- Teaching assistant (2018-2020)-Delivered Electricity and Magnetism laboratory course for undergraduate students and evaluated and graded examinations.
- Presented research at ISMRM (500+ attendees).

Project: Partial Volume (PV) Quantification of Magnetic Resonance Fingerprint (MRF) data.

- Developed a fast and interpretable technique for PV quantification using data transformation techniques, machine learning, and Bayesian analysis, with applications to tissue classification of cerebral gliomas.

Project: Single Half Cylinder B0 Shim Coil for the Knee.

- Simulated and prototyped a half-cylinder B0 shim coil for reducing artifacts in knee MRI scans. In-vivo experimentation achieved close to 70% artefact reduction in regions of interest.

Complex Systems Research Group (SLIIT) Sri Lanka

Visiting Researcher, 2017-2018

- Mentored student researchers to complete projects in computer science and robotics.
- Designed and delivered lecture for undergraduate students and evaluated and graded examinations.
- Presented research at the International Conference on Computational Modelling and Simulation, Colombo

Project: Framework for Quantifying the Resilience of Complex Networks.

- Designed and developed an interpretable and generalized data analysis framework for quantifying the robustness of complex networks, with applications to three real-world networks.

TECHNICAL SKILLS

Machine Learning, Deep Learning, Computer Vision, Statistical Modeling, Data Visualization, Bayesian Computing, Blind Source Separation, Complex Systems, Finite Element Methods, Functional Optimization.

Matlab, Python, Tensorflow, OpenCV, R, NetworkX, C++, OpenGL, SQL, Linux, Shell Scripting, HPC, GPU Computing, Git, MIM, Pinnacle, 3DSlicer.

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LABORATORY EXPERIENCE

Graduate Physics Lab

- Independent experimentation with research grade instrumentation (TDS3000B Series Digital Phosphor Oscilloscope, Keithley Model 199 System DMM Scanner, BK Precision Model 4011 5 MHz Function Generator), computer aided DAQ and control, and digital signal simulation and analysis.

Robotics Lab.

- Designed and implemented embedded systems and joint motion control for an obstacle-avoiding robot.

Computer Graphics Lab

- Developed a 3D scene (using OpenGL) that included a room with a glass cube, a stack of books on a table, and multiple reflections and shadows cast by a waning and waxing sun.

Computational Physics Lab

- Developed mastery in computational skills and foundational algorithms, including integrals and derivatives, linear and nonlinear equations, ODE, PDE, etc.

Computational Statistical Mechanics Lab

- Utilized an algorithmic approach to modern physics, involving techniques such as Monte Carlo simulations, density matrices, path integrals, and simulated annealing.

PUBLICATIONS

- S Weragoda, et al." Deep-Learning Based Liver Dome Delineation on KV Planar Images for Automatic Liver Dome Gating". AAPM 2023.
- B, Guo, S Weragoda, et al." Deep learning-based lung tumor and ipsilateral diaphragm segmentation on kV triggered images for marker less gating of lung SBRT". AAPM 2023.
- Y Chen, R Boyacioglu, S Weragoda et al." Tissue Classification of Cerebral Gliomas using MRF Signal and Deep Learning". ISMRM 2022.
- Y Chen, R Boyacioglu, S Weragoda et al." Improving Annotation Accuracy in MRI data using MRF and Deep Learning". ISMRM 2022.
- M Fan, S Weragoda et al." Single Half-Cylinder B0 Shim Coil for the Knee". ISMRM 2021.
- P Ratnayake, S Weragoda et al. "Quantifying the Robustness of Complex Networks with Heterogeneous Nodes". Mathematics 2021.
- M Bandara, S Weragoda, et al." Overlay Community detection using Community Networks". IEEE SSCI 2018.
- S Weragoda et al. "Resilience of Complex Networks with Heterogeneous Nodes". International Conference on Computational Modelling and Simulation 2017. Colombo.

LEADERSHIP AND BUSINESS SKILLS

- Vice-President of Sri Lankan Student Association. CWRU (2021-2023).

In response to the challenges posed by the pandemic, I founded the association with the aim of creating a nurturing environment for the Sri Lankan student community. I led a diverse group of student officers in organizing various cultural events and offering support to incoming Sri Lankan students, ensuring they feel at home.

- Weatherhead Executive Education Workshop 2023-Developing Power and Influence for Women in Leadership.

I was honored to receive a competitive training grant, in recognition of my leadership role as a woman of color on campus.

- Reviewer and Judge, SOURCE Undergraduate Research Symposium. CWRU (2021-2023).

- Physics Olympiad Organizing Committee UOC (2014-2016)

Worked in a diverse team of 30 officers to organize multiple outreach events and workshops for rural high-schools, which resulted in an almost 30% increase in rural student participation.

- Local Committee Vice President -AIESEC UOC (2014-2016)

Led a diverse student team and collaborated with local grassroots organizations to create positive social impact in areas of education and women empowerment.

- Six-time national winner of the school level English creative writing contest, Sri Lanka

- Commendation award at The Queen's Commonwealth Essay Competition.

- Languages- English (Bilingual proficiency)
Sinhala (Bilingual proficiency)
French (Elementary proficiency)